# Coupler Curves of Planar Four-Bar Linkages

Detailed Notes based on:

(1) *Coupler Curves of Planar Four-Bar Linkages* – J. Angeles

(2) *Finding Straight Line Generators through the Approximate Synthesis of Symmetric Four-Bar Coupler Curves* – Baskar, Plecnik & Hauenstein (2023)

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## 1 Setup and Terminology

### 1.1 The Four-Bar Linkage

A planar four-bar linkage consists of four rigid links connected by four revolute (pin) joints, forming a closed kinematic chain. The links are:

1. **Frame** (ground link, length $a_1$): the fixed link connecting pivots $O_1$ and $O_4$.
2. **Crank** (input link, length $a_2$): rotates about the fixed pivot $O_1$.
3. **Coupler** (floating link, length $a_3$): connects the free ends of the crank and follower. This is the link of primary interest.
4. **Follower** (output link, length $a_4$): connects the free end of the coupler to the fixed pivot $O_4$.

The free end of the crank is $O_2$, and the free end of the follower is $O_3$, so the coupler connects $O_2$ to $O_3$. The entire mechanism has **one degree of freedom** (by Grübler's formula: $3(4-1)-2(4)=1$), so specifying the crank angle $\theta$ completely determines the configuration.

### 1.2 The Coupler Frame and the Tracer Point

Fix a Cartesian frame $\mathcal{F}$ with origin at $O_1$ and axes $X, Y$. Since the coupler is a rigid body, we can attach a second frame $\mathcal{G}$ to it, with origin at the tracer point $P(x, y)$ and axes $U, V$ parallel to a fixed direction only when the coupler is in its reference configuration.

The tracer point $P$ is any point rigidly attached to the coupler link – not necessarily lying on the segment $O_2O_3$. As the crank rotates through a full revolution (if the Grashof condition permits), $P$ traces a closed curve in the plane. This is the **coupler curve**.

### 1.3 Geometric Setup for the Derivation

The position vectors of $O_2$ and $O_3$ from $P$, expressed in $\mathcal{G}$, are constant (since $P$ is rigidly attached to the coupler):

$$\left[\overrightarrow{PO_2}\right]_{\mathcal{G}} = \begin{pmatrix} -u \\ -v \end{pmatrix} = \text{const}, \tag{1}$$

$$\left[\overrightarrow{PO_3}\right]_{\mathcal{G}} = \begin{pmatrix} w \\ -v \end{pmatrix} = \text{const}. \tag{2}$$

Here $u, v, w$ are fixed real numbers that specify where $P$ sits on the coupler triangle. The position of $P$ in $\mathcal{F}$ is simply:

$$\left[\overrightarrow{O_1P}\right]_{\mathcal{F}} = \begin{pmatrix} x \\ y \end{pmatrix}. \tag{3}$$

**Remark 1.1.** The fact that $v$ appears in both expressions (with a minus sign) is a specific geometric convention: $P$, $O_2$, and $O_3$ are taken to have the same $V$-coordinate in $\mathcal{G}$, i.e. $P$ lies on a line parallel to the coupler base $O_2O_3$.

The coupler makes an angle $\theta$ with the $X$-axis. The rotation matrix from $\mathcal{F}$ to $\mathcal{G}$ is:

$$Q = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}. \tag{4}$$

# 2 Derivation of the Coupler Curve Equation

## 2.1 Two Constraint Equations from Link Lengths

Since $O_1O_2$ has fixed length $a_2$ and $O_4O_3$ has fixed length $a_4$, we have:

$$\left\|\overrightarrow{O_1O_2}\right\|^2 = a_2^2, \qquad \left\|\overrightarrow{O_4O_3}\right\|^2 = a_4^2. \tag{5}$$

#### 2.1.1 First Constraint: the Crank

Using the vector addition $\overrightarrow{O_1O_2} = \overrightarrow{O_1P} + \overrightarrow{PO_2}$, we need to express $\overrightarrow{PO_2}$ in $\mathcal{F}$. Since it is known in $\mathcal{G}$, we apply the rotation $Q$:

$$\left[\overrightarrow{PO_2}\right]_{\mathcal{F}} = Q\left[\overrightarrow{PO_2}\right]_{\mathcal{G}} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}\begin{pmatrix} -u \\ -v \end{pmatrix} = \begin{pmatrix} -u\cos\theta + v\sin\theta \\ -u\sin\theta - v\cos\theta \end{pmatrix}. \tag{6}$$

Therefore:

$$\left[\overrightarrow{O_1O_2}\right]_{\mathcal{F}} = \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} -u\cos\theta + v\sin\theta \\ -u\sin\theta - v\cos\theta \end{pmatrix} = \begin{pmatrix} x - u\cos\theta + v\sin\theta \\ y - u\sin\theta - v\cos\theta \end{pmatrix}. \tag{7}$$

Substituting into $\|\overrightarrow{O_1O_2}\|^2 = a_2^2$ and expanding:

$$\begin{aligned} &(x - u\cos\theta + v\sin\theta)^2 + (y - u\sin\theta - v\cos\theta)^2 = a_2^2 \\ &x^2 - 2x(u\cos\theta - v\sin\theta) + (u\cos\theta - v\sin\theta)^2 \\ &\quad + y^2 - 2y(u\sin\theta + v\cos\theta) + (u\sin\theta + v\cos\theta)^2 = a_2^2. \end{aligned} \tag{8}$$

Note that $(u\cos\theta - v\sin\theta)^2 + (u\sin\theta + v\cos\theta)^2 = u^2 + v^2$ (since $Q$ is orthogonal and preserves norms). Expanding the cross terms:

$$-2x(u\cos\theta - v\sin\theta) - 2y(u\sin\theta + v\cos\theta) = -2(ux + vy)\cos\theta + 2(vx - uy)\sin\theta.$$

So the first constraint equation becomes:

$$\boxed{x^2 + y^2 - 2(ux + vy)\cos\theta + 2(vx - uy)\sin\theta + u^2 + v^2 - a_2^2 = 0.} \tag{10a}$$

### 2.1.2 Second Constraint: the Follower

The vector $\overrightarrow{O_4O_3}$ satisfies:

$$\overrightarrow{O_4O_3} = \overrightarrow{O_1O_3} - \overrightarrow{O_1O_4} = \overrightarrow{O_1P} + \overrightarrow{PO_3} - \overrightarrow{O_1O_4}. \tag{9}$$

Now $\overrightarrow{O_1O_4} = (a_1, 0)^T$ (since $O_4$ is at distance $a_1$ along the $X$-axis from $O_1$), so:

$$\left[\overrightarrow{O_1P} - \overrightarrow{O_1O_4}\right]_{\mathcal{F}} = \begin{pmatrix} x - a_1 \\ y \end{pmatrix}. \tag{10}$$

Transforming $\overrightarrow{PO_3}$ into $\mathcal{F}$:

$$\left[\overrightarrow{PO_3}\right]_{\mathcal{F}} = Q\begin{pmatrix} w \\ -v \end{pmatrix} = \begin{pmatrix} w\cos\theta + v\sin\theta \\ w\sin\theta - v\cos\theta \end{pmatrix}. \tag{11}$$

Therefore:

$$\left[\overrightarrow{O_4O_3}\right]_{\mathcal{F}} = \begin{pmatrix} x - a_1 + w\cos\theta + v\sin\theta \\ y + w\sin\theta - v\cos\theta \end{pmatrix}. \tag{12}$$

Substituting into $\|\overrightarrow{O_4O_3}\|^2 = a_4^2$ and expanding (the calculation mirrors the first constraint with $u \to -w$, $a_1$ shift on $x$, $a_2 \to a_4$):

$$\begin{aligned}(x - a_1)^2 + 2(x - a_1)(w\cos\theta + v\sin\theta) + (w\cos\theta + v\sin\theta)^2 \\ + y^2 + 2y(w\sin\theta - v\cos\theta) + (w\sin\theta - v\cos\theta)^2 = a_4^2.\end{aligned} \tag{13}$$

Again $(w\cos\theta + v\sin\theta)^2 + (w\sin\theta - v\cos\theta)^2 = w^2 + v^2$. The cross terms give:

$$2(x - a_1)(w\cos\theta + v\sin\theta) + 2y(w\sin\theta - v\cos\theta) = 2[w(x - a_1) - vy]\cos\theta + 2[v(x - a_1) + wy]\sin\theta.$$

Expanding $(x - a_1)^2 = x^2 - 2a_1x + a_1^2$ and collecting:

$$\boxed{x^2 + y^2 + 2[w(x - a_1) - vy]\cos\theta + 2[v(x - a_1) + wy]\sin\theta - 2a_1x + a_1^2 + v^2 + w^2 - a_4^2 = 0.} \tag{10b}$$

## 2.2 Eliminating One Trigonometric Term by Subtraction

Both (10a) and (10b) are quadratic in $x, y$ and linear in $\cos\theta$ and $\sin\theta$. Subtracting (10b) from (10a):

$$\begin{aligned}&[-2(ux + vy) - 2w(x - a_1) + 2vy]\cos\theta \\ &\quad + [2(vx - uy) - 2v(x - a_1) - 2wy]\sin\theta \\ &\quad + (u^2 - w^2) - (a_2^2 - a_4^2) + 2a_1x - a_1^2 = 0.\end{aligned} \tag{14}$$

Let us simplify the $\cos\theta$ coefficient:

$$-2ux - 2vy - 2wx + 2a_1w + 2vy = -2(u + w)x + 2a_1w.$$

And the $\sin\theta$ coefficient:

$$2vx - 2uy - 2vx + 2a_1v - 2wy = 2a_1v - 2(u + w)y + 2a_1v.$$

Wait – let us be more careful. The $\sin\theta$ coefficient is:

$$2(vx-uy)-2v(x-a_1)-2wy = 2vx-2uy-2vx+2a_1v-2wy = -2uy-2wy+2a_1v = -2(u+w)y+2a_1v.$$

So (10a) − (10b) gives:

$$\boxed{2[(u + w)x - a_1w]\cos\theta + 2[2a_1v - (u + w)y]\sin\theta - 2a_1x + a_1^2 + a_2^2 - a_4^2 + u^2 + w^2 = 0.} \tag{10c}$$

**Why this step matters.** Equation (10c) is now *linear* in $x$ and $y$ (not quadratic), while still being linear in $\cos\theta$ and $\sin\theta$. This is crucial: it means when we later form the $4 \times 4$ determinant, two of the rows will be linear in $x, y$ (from (10c) and its multiplied version) rather than quadratic. This is exactly why the resulting implicit curve has degree $2 + 2 + 1 + 1 = 6$ rather than higher.

## 2.3 The Weierstrass / Tan-Half-Angle Substitution

To eliminate $\theta$ algebraically, we use the substitution $T = \tan(\theta/2)$, which gives:

$$\cos\theta = \frac{1 - T^2}{1 + T^2}, \qquad \sin\theta = \frac{2T}{1 + T^2}. \tag{11}$$

This transforms any equation that is rational in $\cos\theta$ and $\sin\theta$ into a rational equation in $T$ alone. After multiplying through by $(1 + T^2)$, any equation linear in $\cos\theta$ and $\sin\theta$ becomes quadratic in $T$.

### 2.3.1 Applying the Substitution to (10b)

Substitute (11) into (10b). Multiplying through by $(1 + T^2)$:

$$(x^2 + y^2)(1 + T^2) + 2[w(x - a_1) - vy](1 - T^2) + 2[v(x - a_1) + wy](2T) \\ + (-2a_1x + a_1^2 + v^2 + w^2 - a_4^2)(1 + T^2) = 0.$$

Collect by powers of $T$:

$$\begin{aligned} T^2 &: \quad (x^2 + y^2) - 2[w(x - a_1) - vy] + (-2a_1x + a_1^2 + v^2 + w^2 - a_4^2) \\ T^1 &: \quad 4[v(x - a_1) + wy] \\ T^0 &: \quad (x^2 + y^2) + 2[w(x - a_1) - vy] + (-2a_1x + a_1^2 + v^2 + w^2 - a_4^2) \end{aligned}$$

Define the coefficients as in the paper:

$$A_1 \equiv x^2 + y^2 + 2(ux + vy) + u^2 - a_2^2 \tag{12c}$$
$$B_1 \equiv 4(vx - uy) \tag{12d}$$
$$C_1 \equiv x^2 + y^2 - 2(ux + vy) + u^2 - a_2^2 \tag{12e}$$

**Remark 2.1.** Note that $A_1$ and $C_1$ differ only in the sign of the $2(ux + vy)$ term. This is a direct consequence of how $\cos\theta$ (which gives $\pm$ terms through $1 \mp T^2$) separates the $T^2$ and $T^0$ coefficients.

Similarly, applying the substitution to (10c):

$$A_2 \equiv -2(a_1 + u + w)x + a_1^2 + a_2^2 + a_4^2 + u^2 + w^2 - 2a_1 w \tag{12f}$$
$$B_2 \equiv 4[-a_1 v + (w + u)y] \tag{12g}$$
$$C_2 \equiv -2(a_1 - u - w)x + a_1^2 + a_2^2 + a_4^2 + u^2 + w^2 + 2a_1 w \tag{12h}$$

So we have the two quadratic-in-$T$ equations:

$$A_1 T^2 - 2B_1 T + C_1 = 0 \tag{12a}$$
$$A_2 T^2 - 2B_2 T + C_2 = 0 \tag{12b}$$

## 2.4 Dialytic Elimination: Building the $4 \times 4$ System

We have two equations (12a) and (12b) in one unknown $T$. To eliminate $T$, we use **dialytic elimination** (also called Sylvester's method or the resultant).

The idea: multiply each equation by $T$ to produce two cubic equations:

$$A_1 T^3 - 2B_1 T^2 + C_1 T = 0 \tag{12i}$$
$$A_2 T^3 - 2B_2 T^2 + C_2 T = 0 \tag{12j}$$

Now we have four equations – (12a), (12b), (12i), (12j) – which are *linear* in the monomials $\{T^3, T^2, T^1, T^0\}$. This can be written as:

$$M\mathbf{x} = \mathbf{0}_4, \tag{13a}$$

where:

$$M = \begin{pmatrix} A_1 & -2B_1 & C_1 & 0 \\ A_2 & -2B_2 & C_2 & 0 \\ 0 & A_1 & -2B_1 & C_1 \\ 0 & A_2 & -2B_2 & C_2 \end{pmatrix}, \qquad \mathbf{x} = \begin{pmatrix} T^3 \\ T^2 \\ T^1 \\ T^0 \end{pmatrix}. \tag{13b}$$

**Note on the typo in the Angeles notes.** The paper writes $\mathbf{x} = (T^4, T^3, T^2, T^1)^T$, which is incorrect. The four equations (12a, 12b, 12i, 12j) involve $T^0$ through $T^3$, so the correct vector is $\mathbf{x} = (T^3, T^2, T^1, T^0)^T$ as written above. One can verify this by checking that each row of $M$ dotted with $\mathbf{x}$ reproduces the corresponding equation.

### 2.4.1 Why the Solution Must Be Non-Trivial

If $T$ is a finite real number, then $\mathbf{x} = (T^3, T^2, T, 1)^T \neq \mathbf{0}$. So a non-trivial solution to $M\mathbf{x} = \mathbf{0}$ exists if and only if $M$ is singular:

$$F(x, y) \equiv \det(M) = 0. \tag{14}$$

This is the implicit equation of the coupler curve.

## 2.5 The Degree Count: Why the Curve is Sextic

**Theorem 2.1.** The coupler curve of a planar four-bar linkage is an algebraic curve of degree 6.

*Proof.* The degree of $\det(M)$ as a polynomial in $x$ and $y$ is determined by the Leibniz expansion:

$$\det(M) = \sum_{\sigma \in S_4} \operatorname{sgn}(\sigma) \prod_{i=1}^{4} M_{i,\sigma(i)}. \tag{15}$$

Each term in the expansion is a product of four entries, one from each row. The entries have the following degrees in $(x, y)$:

- Rows 1 and 3 contain entries from (12a)/(12i): $A_1, B_1, C_1$ are **degree 2** in $(x, y)$, and 0 (degree $-\infty$, contributes 0).
- Rows 2 and 4 contain entries from (12b)/(12j): $A_2, B_2, C_2$ are **degree 1** in $(x, y)$, and 0.

The highest-degree term comes from choosing the non-zero entry from each row that maximises the total degree. One such choice is:

$$M_{1,1} \cdot M_{2,2} \cdot M_{3,3} \cdot M_{4,4} = A_1 \cdot (-2B_2) \cdot (-2B_1) \cdot C_2,$$

which has degree $2 + 1 + 2 + 1 = 6$.

No product of four entries (one per row) can exceed degree $2 + 1 + 2 + 1 = 6$, since rows 1,3 contribute at most degree 2 each and rows 2,4 at most degree 1 each. Therefore $\det(M)$ is a polynomial of degree exactly 6 in $(x, y)$. □

**Corollary 2.1.** The coupler curve cannot contain a straight segment of finite length.

*Proof.* By Bézout's theorem, a degree-6 algebraic curve $F(x, y) = 0$ and a line (degree-1 curve) intersect in at most $6 \times 1 = 6$ points (counted with multiplicity, in projective space). A straight segment of finite length, if it lay on the coupler curve, would supply infinitely many intersection points with the line containing it – a contradiction. Therefore no such segment can exist. □

**Remark 2.2.** This result is physically important: the "straight-line" linkages of Watt, Roberts, Chebyshev, and Evans do not trace exact straight lines. They trace sextic curves that are highly flattened in a certain region. The design problem is to find link parameters that make this flattening as close to perfect as possible over the operating range.

# 3 Isotropic Coordinate Formulation (Baskar et al. 2023)

The Baskar et al. paper reformulates everything using **isotropic (complex) coordinates**, which offers algebraic advantages for numerical continuation methods. We now develop this framework carefully.

## 3.1 Isotropic Coordinates

In isotropic coordinates, a point $P = (p, q) \in \mathbb{R}^2$ is represented as a pair $(P, P^*)$ where:

$$P = p + iq, \qquad P^* = p - iq. \tag{16}$$

Here $P^*$ is the complex conjugate of $P$, and $i = \sqrt{-1}$. This is simply the identification of the Euclidean plane $\mathbb{R}^2$ with the complex plane $\mathbb{C}$.

**Why isotropic coordinates?**

1. Rotations become multiplications: rotating by angle $\phi$ maps $P \mapsto e^{i\phi}P$.
2. The distance between points $P$ and $Q$ satisfies $|P - Q|^2 = (P - Q)(P^* - Q^*)$.
3. Polynomial resultant computations become cleaner when rotation operators appear as multiplicative factors.
4. The coupler trace equation factors more naturally, enabling efficient numerical homotopy continuation.

## 3.2 Design Variables

The four-bar linkage is described by the design vector:

$$\boldsymbol{d} = \{A, A^*, B, B^*, l_1, l_2, l_3, Q, Q^*\}, \tag{17}$$

where:

- $A, A^* \in \mathbb{C}$: isotropic coordinates of the fixed pivot $O_1$ (left ground pivot).
- $B, B^* \in \mathbb{C}$: isotropic coordinates of the fixed pivot $O_4$ (right ground pivot).
- $l_1, l_2, l_3 \in \mathbb{R}_{>0}$: lengths of the crank, coupler base, and follower respectively.
- $Q, Q^* \in \mathbb{C}$: the coupler trace point in the local frame of the coupler, normalised by the coupler base length $l_2$. Explicitly, $Q$ is a *stretch-rotation* that maps a vector of real length $l_2$ in the coupler frame to the complex coordinates of the trace point. Note that $Q^*$ is the complex conjugate of $Q$.

The trace point in the global frame is $X$ (with conjugate $X^*$), and this is the locus we seek.

## 3.3 Rotation Operators

Define:

$$\Phi_k = e^{i\phi_k}, \quad k = 1, 2, 3, \tag{18}$$

where $\phi_k$ is the angular displacement of the $k$-th moving link (crank, coupler, follower respectively) measured counter-clockwise from the positive $x$-axis.

Since $\Phi_k$ lies on the unit circle in $\mathbb{C}$, its inverse (conjugate) is:

$$\Phi_k^{-1} = \overline{\Phi_k} = e^{-i\phi_k} = \frac{1}{\Phi_k}. \tag{19}$$

## 3.4 Vector Loop Equations

The closure of the four-bar linkage imposes two vector loop equations. These express that the trace point $X$ can be reached from either ground pivot:

**Left dyad** (from pivot $A$ through crank $l_1$ and partial coupler $l_2Q$):

$$A + l_1\Phi_1 + l_2Q\Phi_2 = X. \tag{1}$$

**Right dyad** (from pivot $B$ through follower $l_3$ and remaining coupler $l_2(Q-1)$):

$$B + l_3\Phi_3 + l_2(Q-1)\Phi_2 = X. \tag{2}$$

**Remark 3.1.** Equation (2) uses $l_2(Q-1)$ because the coupler joint $O_3$ corresponds to the unit displacement $\Phi_2$ along the coupler base, so the vector from $O_3$ to the trace point $X$ is $l_2(Q-1)\Phi_2$. The term $Q-1$ captures the offset from the right coupler pin to the trace point.

Taking complex conjugates (which correspond to the same geometric equations expressed in terms of the conjugate coordinates):

$$A^* + l_1\frac{1}{\Phi_1} + l_2Q^*\frac{1}{\Phi_2} = X^*, \tag{3}$$

$$B^* + l_3\frac{1}{\Phi_3} + l_2(Q^*-1)\frac{1}{\Phi_2} = X^*. \tag{4}$$

Together, equations (1)–(4) form a system of 4 complex equations in the 3 unknowns $\Phi_1, \Phi_2, \Phi_3$ (plus the known design variables $\boldsymbol{d}$ and the traced point $X, X^*$).

## 3.5 Eliminating the Rotation Operators: the Coupler Trace Equation

The rotation operators $\Phi_1, \Phi_2, \Phi_3$ are not design specifications – they are the kinematic variables that describe the current posture of the linkage. We want to eliminate them to obtain a single implicit equation $f(X, X^*; \boldsymbol{d}) = 0$ that the trace point must satisfy for *all* postures.

### 3.5.1 Strategy

From equations (1) and (3), we can express $\Phi_1$ and $1/\Phi_1$ in terms of $X, \Phi_2, \boldsymbol{d}$:

$$l_1\Phi_1 = X - A - l_2Q\Phi_2, \qquad l_1\frac{1}{\Phi_1} = X^* - A^* - l_2Q^*\frac{1}{\Phi_2}. \tag{20}$$

Since $\Phi_1$ lies on the unit circle, $|\Phi_1|^2 = \Phi_1 \cdot (1/\Phi_1) = 1$, which gives the constraint:

$$l_1^2 = (X - A - l_2Q\Phi_2)(X^* - A^* - l_2Q^*/\Phi_2). \tag{21}$$

Similarly eliminating $\Phi_3$ from (2) and (4). Then eliminating $\Phi_2$ from the two resulting equations, the authors obtain the scalar coupler trace equation via polynomial resultants, arriving at the $4 \times 4$ determinant:

$$
\begin{vmatrix}
Q^*(A-X) & g(X,X^*;\boldsymbol{d}) & l_2Q(A^*-X^*) & 0 \\
0 & l_2Q^*(A-X) & g(X,X^*;\boldsymbol{d}) & Q(A^*-X^*) \\
(Q^*-1)(B-X) & h(X,X^*;\boldsymbol{d}) & l_2(Q-1)(B^*-X^*) & 0 \\
0 & l_2(Q^*-1)(B-X) & h(X,X^*;\boldsymbol{d}) & (Q-1)(B^*-X^*)
\end{vmatrix} = 0, \tag{5}
$$

where:

$$
g(X,X^*;\boldsymbol{d}) = l_2^2QQ^* - l_1^2 + (A-X)(A^*-X^*), \tag{22}
$$
$$
h(X,X^*;\boldsymbol{d}) = l_2^2(Q-1)(Q^*-1) - l_3^2 + (B-X)(B^*-X^*). \tag{23}
$$

**Physical meaning of $g$ and $h$:**

- $g = l_2^2QQ^* - l_1^2 + |A-X|^2$: This is a measure of how well the crank constraint is satisfied. It involves the crank length $l_1$, the trace point offset $QQ^* = |Q|^2$, and the distance $|A-X|^2$ from the left pivot to the trace point.
- $h = l_2^2(Q-1)(Q^*-1) - l_3^2 + |B-X|^2$: Similarly for the follower, with the offset now $|Q-1|^2$.

### 3.6 The Sextic and Its Monomial Structure

Expanding determinant (5), the coupler trace equation $f(X,X^*;\boldsymbol{d}) = 0$ is a degree-6 polynomial in $(X,X^*)$. It is well-known that this equation has **circularity 3**, meaning it passes through each of the two circular points at infinity $I = (1:i:0)$ and $\bar{I} = (1:-i:0)$ with multiplicity 3. This restricts the monomial basis to exactly 16 terms:

$$
\{X^3X^{*3},\ X^3X^{*2},\ X^3X^*,\ X^3,\ X^2X^{*3},\ X^2X^{*2},\ X^2X^*,\ X^2,\ XX^{*3},\ XX^{*2},\ XX^*,\ X,\ X^{*3},\ X^{*2},\ X^*,\ 1\}. \tag{24}
$$

The leading coefficient (of $X^3X^{*3}$) is exactly 1. The remaining 15 coefficients are polynomial functions of the design variables $\boldsymbol{d}$, as listed in the Appendix of Baskar et al.

**Remark 3.2** (On circularity)**.** A general degree-6 polynomial in $(x,y)$ would have $\binom{8}{2} =$ 28 monomials. The circularity-3 constraint (each circular point at infinity is a triple root) reduces this to 16. Geometrically, this reflects the fact that the coupler curve is a *tricircular sextic* – a classical object in 19th-century kinematics.

# 4 Roberts Cognate Triplets

**Theorem 4.1** (Roberts–Chebyshev Theorem)**.** *Every coupler curve of a four-bar linkage is generated by exactly three different four-bar linkages, called* Roberts cognates*.*

For a design $\boldsymbol{d}_1 = \{A, A^*, B, B^*, l_1, l_2, l_3, Q, Q^*\}$, the other two cognates are:

$$\boldsymbol{d}_2 = \Big\{B,\ B^*,\ A + Q(B - A),\ A^* + Q^*(B^* - A^*),\ l_2\sqrt{(1-Q)(1-Q^*)},$$
$$l_3\sqrt{(1-Q)(1-Q^*)},\ l_1\sqrt{(1-Q)(1-Q^*)},\ \frac{1}{1-Q},\ \frac{1}{1-Q^*}\Big\}, \tag{6a}$$

$$\boldsymbol{d}_3 = \Big\{A + Q(B - A),\ A^* + Q^*(B^* - A^*),\ A,\ A^*,\ l_3\sqrt{QQ^*},\ l_1\sqrt{QQ^*},\ l_2\sqrt{QQ^*},\ \frac{Q-1}{Q},\ \frac{Q^*-1}{Q^*}\Big\}. \tag{6b}$$

**Remark 4.1.** These transformations are not arbitrary – they correspond to forming similar triangles on each link of the original linkage, a geometric construction that preserves the coupler curve while permuting the roles of crank, coupler, and follower.

# 5 Symmetric Coupler Curves

## 5.1 Motivation

The Baskar et al. paper restricts to four-bars that generate **symmetric coupler curves** – curves that are mirror-symmetric about some axis in the plane. This restriction is motivated by two observations:

1. The classical straight-line generators (Watt, Evans, Roberts, Chebyshev) all produce symmetric coupler curves.
2. Symmetry imposes algebraic conditions on $\boldsymbol{d}$ that reduce the 9-dimensional design space to 7 dimensions (and further to 3 when ground pivots are fixed), making the optimisation tractable.

## 5.2 Axis of Symmetry in Isotropic Coordinates

A generic line in the plane (the axis of symmetry) is described in isotropic coordinates as:

$$L(P, P^*) = K^*P + KP^* + c = 0, \tag{7}$$

where $K \in \mathbb{C}$ is a vector along the normal to the line, and $c \in \mathbb{R}$ is determined by any point $D$ on the line via $c = -K^*D - KD^*$.

The reflection of a point $(X, X^*)$ about this axis is:

$$(X_m, X_m^*) = \left(-\frac{c + KX^*}{K^*},\ -\frac{c + K^*X}{K}\right). \tag{8}$$

**Derivation of eq. (8).** The reflection of $X$ about the line $K^*P + KP^* + c = 0$ maps $P$ to $P_m$ such that the midpoint $\frac{P+P_m}{2}$ lies on the line and $P_m - P$ is parallel to $K$ (the normal direction).
Writing $P_m = X_m$, the midpoint condition gives $K^*\frac{X+X_m}{2} + K\frac{X^*+X_m^*}{2} + c = 0$, i.e.:

$$K^*X_m + KX_m^* = -K^*X - KX^* - 2c.$$

The normality condition means $X_m - X = \lambda K$ for some real $\lambda$, so $X_m^* - X^* = \lambda K^*$.

Substituting and solving:

$$\lambda = -\frac{K^* X + K X^* + 2c}{2|K|^2},$$

giving $X_m = X + \lambda K = -\frac{c+KX^*}{K^*}$ (using $|K|^2 = KK^*$). The expression for $X_m^*$ follows by conjugation.

### 5.3 Condition for a Symmetric Coupler Curve

For the coupler curve to be symmetric about the axis $L(P, P^*) = 0$, the reflected point $(X_m, X_m^*)$ must also lie on the coupler curve whenever $(X, X^*)$ does. This means:

$$f(X_m, X_m^*; \boldsymbol{d}) = \lambda f(X, X^*; \boldsymbol{d}), \tag{9}$$

for some scalar $\lambda$. The constant $\lambda = 1/(K^3 K^{*3})$ appears because of the $K, K^*$ factors in the denominators of $X_m, X_m^*$, which shift the degree when substituted into the degree-6 equation.

Substituting the reflection (8) into the 16-term sextic and equating coefficients monomial-by-monomial with the right-hand side yields 15 conditions (the leading $X^3 X^{*3}$ term is normalised to 1, so it is automatically satisfied) on the design variables $\boldsymbol{d}$ and the axis parameters $K, K^*, c$.

### 5.4 The Three Simplest Symmetry Conditions

The conditions corresponding to the monomials $X^3$, $X^3 X^*$, $X^3 X^{*2}$ (which are the easiest to extract) give:

$$c(c+K)(c+KQ^*) = 0, \tag{10}$$

$$-3c^2 - 2cK^* - 2cK^*Q - K^{*2}Q + K^2Q^* = 0, \tag{11}$$

$$3c + K + K^* + K^*Q + KQ^* = 0. \tag{12}$$

Equation (10) immediately tells us $c = 0$, $c = -K$, or $c = -KQ^*$. Each branch leads to a different class of linkage.

### 5.5 Classification: Class A and Class B

#### 5.5.1 Case 1: $c = 0$

With $c = 0$, eqs. (11) and (12) become:

$$-K^{*2}Q + K^2Q^* = 0, \tag{13}$$

$$K + K^* + K^*Q + KQ^* = 0. \tag{14}$$

This is a linear system in $Q, Q^*$. Two sub-cases arise depending on its rank:

**Sub-case: Rank-deficient system.** The system is rank-deficient if $K = -K^*$ (i.e. $K$ is purely imaginary) and $Q = Q^*$ (i.e. $Q$ is real). The 15 conditions all vanish identically, meaning *any* link lengths are allowed. These are **Class A** linkages: the trace point lies along the coupler base $O_2O_3$, and the coupler curve is symmetric about the ground link by a simple reflection argument.

**Sub-case: Well-determined system.** If the system has full rank, solving gives $Q = -K/K^*$ and $Q^* = -K^*/K$. For all 15 conditions to be satisfied, one also needs $l_1 = l_2$. These are **Class B** sub-class 1: $Q = 1/Q^*$ with $l_1 = l_2$.

#### 5.5.2 Case 2: $c = -K = -K^*$

Here $K = K^*$ (real), so the axis of symmetry is horizontal (or vertical, depending on normalisation). Equations (11) and (12) give $Q^* = 1 - Q$ and $l_1 = l_3$. This is **Class B sub-class 2**.

#### 5.5.3 Case 3: $c = -KQ^* = -K^*Q$

Similar analysis gives $Q^* = Q/(Q-1)$ and $l_2 = l_3$. This is **Class B sub-class 3**.

### 5.6 Summary

**Class A**: $Q = Q^*$ (trace point on coupler base). No restriction on link lengths. Symmetric about the ground link.
**Class B**: Three sub-classes constituting a Roberts cognate triplet:

1. $Q = 1/Q^*$, $l_1 = l_2$
2. $Q^* = 1 - Q$, $l_1 = l_3$ $\quad\leftarrow$ *This is the Roberts linkage sub-class*
3. $Q^* = Q/(Q-1)$, $l_2 = l_3$

The axis of symmetry for Class B is the perpendicular bisector of the fixed link (ground link of cognate #2). All three cognates share the same axis of symmetry.

## 6 Optimisation Formulation for Straight-Line Synthesis

### 6.1 The Design Problem

We want to find the four-bar linkage (within Class B, sub-class 2) whose coupler curve most closely approximates a desired straight-line segment. The design variables, after imposing $Q^* = 1 - Q$ and $l_1 = l_3 = l$, are reduced as follows.

#### 6.1.1 Reduction of Variables

Since $Q + Q^* = 1$, we can write:

$$Q = \frac{1}{2} + iq_y, \tag{25}$$

where $q_y \in \mathbb{R}$ is the single free parameter describing the trace point position. Note $Q^* = \frac{1}{2} - iq_y$.

The link lengths $l$ and $l_2$ appear only as $l^2$ and $l_2^2$ in the coupler trace equation (since all constraints are of the form $|\cdot|^2 = \text{link length}^2$). Define:

$$l_s = l^2 - \frac{l_2^2}{4}(1 + 4q_y^2), \qquad l_{2s} = l_2^2. \tag{26}$$

The coupler trace equation simplifies into a polynomial in the 3 real design variables $\{l_s, l_{2s}, q_y\}$, with the ground pivot locations $A, A^*, B, B^*$ treated as **fixed parameters**.

This reduces the design search from 9 variables to **3 variables**, for a given choice of ground pivots.

## 6.2 The Objective Function

Given $N$ desired design positions $\{X_j, X_j^*\}_{j=1}^N$ on the target straight line, the coupler equation evaluated at these points gives residuals:

$$\eta_j = f(X_j, X_j^*;\ A, A^*, B, B^*,\ l_s, l_{2s}, q_y). \tag{27}$$

The optimisation objective is the $L^2$ norm of these residuals:

$$\min_{l_s, l_{2s}, q_y} \sum_{j=1}^{N} \eta_j^2. \tag{15}$$

### 6.2.1 Continuous Limit

As $N \to \infty$ with the design positions densely sampling a continuous parametric curve $X(t), X^*(t)$ for $t \in [t_i, t_f]$, the objective becomes the integral:

$$\xi = \int_{t_i}^{t_f} \eta(t)^2\, dt, \tag{16}$$

where $\eta(t) = f(X(t), X^*(t);\ \boldsymbol{d})$.

**Why $L^2$?** The authors choose the $L^2$ norm specifically to preserve the polynomial structure of the objective in the design variables $l_s, l_{2s}, q_y$. The integrand $\eta^2$ is a polynomial in these variables (of degree 55 when fully expanded), with coefficients that are numerical integrals (the moments of the desired curve). This polynomial structure is essential for applying homotopy continuation.

## 6.3 Moments of the Desired Curve

Since $f$ is a degree-6 polynomial in $X, X^*$, the integrand $\eta^2$ is degree 12 in $X, X^*$ and degree $\leq 4$ in each of $l_s, l_{2s}, q_y$ (by the structure of the coupler trace equation after specialisation). The coefficients that appear are:

$$M_{j,k} = \int_{t_i}^{t_f} X(t)^j X^*(t)^k\, dt, \quad 0 \leq j, k \leq 6. \tag{28}$$

There are $7 \times 7 = 49$ such moments (though not all appear independently). These are purely numerical integrals over the desired curve – they can be computed once and stored, making the optimisation fast for any desired trace.

## 6.4 First-Order Optimality Conditions

The critical points of $\xi$ satisfy:

$$\frac{\partial \xi}{\partial l_s} = 0, \qquad \frac{\partial \xi}{\partial l_{2s}} = 0, \qquad \frac{\partial \xi}{\partial q_y} = 0. \tag{17}$$

This is a system of 3 polynomial equations in 3 unknowns. The degrees are:

- $\partial\xi/\partial l_s$: degree $\leq 3$ in $(l_s, l_{2s}, q_y)$, degree $\leq 2$ in $l_{2s}$ and $q_y$ individually.
- Similarly for the other two equations.

The total degree of this polynomial system is 648 (the naive Bézout bound). However:

- The **2-homogeneous Bézout bound** is 186 (exploiting the bi-graded structure of the system).
- The **BKK bound** (Bernstein-Kushnirenko-Khovanskii, based on mixed volumes of Newton polytopes) is **73**.

The authors confirm that 73 is tight by solving a generic system with the continuation solver `Bertini` and finding exactly 73 finite solutions.

**Significance of the BKK bound of 73.** This means: for any given choice of ground pivot locations $A, B$ and any desired trace curve, the optimisation problem has *at most 73 critical points*. Homotopy continuation can find all of them by tracking 73 solution paths. The computation takes about 15 seconds per ground pivot choice, making a systematic grid search feasible.

## 6.5 Homotopy Continuation Solution

The authors solve the system (17) using **parameter homotopy**. The idea is:

1. Solve a *start system* (a generic instance of the polynomial system) to find 73 solutions. This is done once using a 2-homogeneous homotopy from 186 start points.
2. For each new choice of ground pivots (i.e. each new parameter value), the 73 start solutions are continuously deformed to the target solutions by tracking solution paths in $\mathbb{C}^3$.
3. Solutions that diverge to infinity are discarded; the finite ones are retained as candidate four-bar designs.

## 6.6 Design of Experiments

The desired curve is the unit segment $X(t) = X^*(t) = t$, $t \in [-0.5, 0.5]$ along the $x$-axis.

Ground pivots are systematically varied across a grid parameterised by $(r, \theta, s)$:

$$\text{Specification I (axis through midpoint):} \quad r \in \{0.25i\}_{i=0}^{8}, \quad \theta \in \{15j\}_{j=0}^{6}, \quad s \in \{0.25k\}_{k=1}^{8}, \tag{18}$$

$$\text{Specification II (axis through endpoint):} \quad r \in \{0.25i\}_{i=0}^{8}, \quad \theta \in \{15j\}_{j=0}^{12}, \quad s \in \{0.25k\}_{k=1}^{8}, \tag{19}$$

giving $9 \times 7 \times 8 + 9 \times 13 \times 8 = 504 + 936 = 1440$ total problems.

The computation yielded 16,904 physical linkages. Filtering for structural error $\leq 1/100$ (straight-line tolerance) and link length ratio $\leq 2$, the final result is **274 distinct approximate straight-line generating four-bars**, which are organised into a 2D atlas using UMAP dimensionality reduction.

# 7 Connection to Seismometer Design

The physical motivation for this project is the design of a suspension mechanism for a seismometer proof mass. The suspension must:

1. Allow the proof mass to move along a **nearly straight line** in response to ground motion.
2. Have **minimal cross-axis coupling**: deviation from the straight-line trajectory introduces a position-dependent restoring force component perpendicular to the intended measurement axis, distorting the instrument's frequency response.
3. Be compact and mechanically simple.

The Roberts linkage (Class B, sub-class 2: $Q^* = 1 - Q$, $l_1 = l_3$) is a natural candidate because its coupler curve has a near-straight central segment, it is symmetric (so the restoring force is symmetric about the rest position), and its cognate structure gives three design options that trace the same curve.

The optimisation framework of Baskar et al. allows systematic search over the 3-dimensional parameter space (for fixed ground pivot geometry) to find the Roberts linkage whose coupler curve most closely approximates a straight line over the seismometer's operating range, subject to the physical constraints of the instrument.

# 8 Mathematica Notebook: `RobertsLinkage.nb`

## 8.1 Overview

The notebook `RobertsLinkage.nb` accompanies these notes and implements the full pipeline from basic four-bar kinematics through to optimisation of the Roberts linkage. It is structured in eleven sections, designed to be run top-to-bottom in Mathematica (version 12+). All functions are defined in earlier sections and called in later ones, so cells should be evaluated in order.

The notebook has two modes of use:

1. **Interactive exploration** (Sections 4, 6): `Manipulate` blocks with sliders for real-time visualisation of coupler curves.
2. **Computational analysis** (Sections 7–11): numerical optimisation, symbolic determinant computation, and deviation profiling.

## 8.2 Section-by-Section Description

### Section 2 – Basic Four-Bar Kinematics

Defines three core functions:

- `FourBarSolve[a1, a2, a3, a4, theta2]`: Given the four link lengths and the crank angle $\theta_2$, solves analytically for the coupler angle $\theta_3$ and follower angle $\theta_4$ using the Freudenstein method. Internally computes the constants:

$$K_1 = \frac{a_1}{a_2}, \quad K_2 = \frac{a_1}{a_4}, \quad K_3 = \frac{a_2^2 - a_3^2 + a_4^2 + a_1^2}{2a_2a_4},$$

  then solves the quadratic $AT^2 + BT + C = 0$ (where $T = \tan(\theta_4/2)$) for $\theta_4$, and similarly for $\theta_3$. Returns `None` if the discriminant is negative (linkage cannot reach that configuration).

- `CouplerPoint[a1, a2, theta2, theta3, u, v]`: Given a configuration $(\theta_2, \theta_3)$ and coupler point coordinates $(u, v)$ in the coupler frame, returns the Cartesian coordinates $(p_x, p_y)$ of the tracer point $P$ in the fixed frame:

$$p_x = a_2 \cos\theta_2 + u\cos\theta_3 - v\sin\theta_3, \quad p_y = a_2 \sin\theta_2 + u\sin\theta_3 + v\cos\theta_3.$$

- `TraceCouplerCurve[a1, a2, a3, a4, u, v, npts]`: Sweeps the crank angle $\theta_2$ over $[0, 2\pi]$ in `npts` steps, calls `FourBarSolve` and `CouplerPoint` at each step, and returns the list of tracer point positions. Skips angles where the linkage cannot assemble.

### Section 3 – Grashof Condition

- `GrashofQ[a1, a2, a3, a4]`: Returns `True` if $\ell_{\min} + \ell_{\max} \le \ell_{\text{mid}_1} + \ell_{\text{mid}_2}$ (Grashof condition for full crank rotation), `False` otherwise.

- `GrashofLabel[...]`: Returns a formatted coloured string for display in graphics.

### Section 4 – Interactive General Four-Bar Explorer

A `Manipulate` block with six sliders:

| Slider | Variable | Range |
|---|---|---|
| Frame | $a_1$ | [0.5, 3.5] |
| Crank | $a_2$ | [0.2, 2.0] |
| Coupler | $a_3$ | [0.5, 3.5] |
| Follower | $a_4$ | [0.2, 2.5] |
| Coupler point $x$ | $u$ | [−1.5, 1.5] |
| Coupler point $y$ | $v$ | [−1.5, 1.5] |

The graphic shows the full coupler curve, a snapshot of the linkage at $\theta_2 = \pi/4$, the fixed pivots, and a Grashof status indicator. The coupler curve is coloured orange for Grashof linkages and blue for non-Grashof. The tracer point is shown as a filled red circle.

### Section 5 – Roberts Symmetry Conditions (Text)

A text cell explaining the reduction of the general design variables to the Roberts subcase:

$$a_2 = a_4 = \ell, \quad u = \frac{a_3}{2}, \quad v = q_y \cdot a_3,$$

leaving three free parameters $\{\ell,\ a_3,\ q_y\}$ for fixed frame $a_1$.

### Section 6 – Roberts Linkage Explorer

Defines two helper functions and a `Manipulate` block specific to the Roberts linkage:

- `RobertsTrace[a1, a2, a3, qy, npts]`: Calls `TraceCouplerCurve` with the Roberts conditions $a_4 = a_2$ and $u = a_3/2$, $v = q_y \cdot a_3$ enforced.
- `StraightLineDeviation[pts]`: Fits a best-fit line $y = mx + b$ to a point set using `Fit[..., {1, x}, x]`, computes residuals, and returns the RMS:

$$\text{RMS} = \sqrt{\frac{1}{N}\sum_{j=1}^{N}(y_j - \hat{y}_j)^2}.$$

- `NearStraightSegment[pts, threshold]`: Approximates local curvature via second differences of the $y$-coordinates. Returns the subset of points where the absolute second difference is below `threshold`, which identifies the near-flat portion of the coupler curve.

The `Manipulate` block has four sliders $(a_1, a_2, a_3, q_y)$ and displays:

- The full coupler curve in grey.
- The near-straight segment highlighted in red.
- The best-fit straight line through the near-straight segment in dashed blue.
- A text overlay showing parameter values and the current RMS deviation.
- A linkage snapshot at $\theta_2 = \pi/3$.

### Section 7 – $L^2$ Deviation Landscape

Computes a grid of RMS deviations over $(a_2, q_y) \in [0.4, 2.0] \times [0.1, 1.0]$ for fixed $a_1 = 2.5$, $a_3 = 2.0$. The grid has spacing $\Delta a_2 = 0.1$ and $\Delta q_y = 0.05$, giving $17 \times 19 = 323$ evaluations. Results are displayed as both a 3D surface plot (`ListPlot3D`) and a contour plot (`ListContourPlot`) coloured by the `"TemperatureMap"` scheme. This is the empirical version of the objective function landscape from eq. (15).

### Section 8 – Optimal Roberts Linkage

- `L2Deviation[a1, a2, a3, qy]`: Computes the RMS deviation for a given parameter set by tracing 600 points, filtering to the near-flat region ($|y| < 1.5$), fitting a line, and returning the residual RMS.
- `OptimiseRoberts[a1, a3]`: Calls `NMinimize` with the `"DifferentialEvolution"` method over $(a_2, q_y) \in (0.2, 2.5) \times (0.05, 1.5)$, returning the optimal parameter values and minimum deviation. Differential evolution is used because the objective landscape may have multiple local minima.

The section then visualises the optimal linkage geometry.

### Section 9 – Family of Coupler Curves

Plots coupler curves for $q_y \in \{0.2, 0.4, 0.6, 0.8, 1.0\}$ (fixed $a_1 = 2.5$, $a_2 = 1.0$, $a_3 = 2.0$), colour-coded by $q_y$ using the `"Rainbow"` colour scheme. This illustrates how the shape of the near-straight segment, and in particular its height above the ground link, varies with the trace point offset.

### Section 10 – Deviation Profile Along the Curve

For a specific Roberts linkage ($a_1 = 2.5$, $a_2 = 1.0$, $a_3 = 2.0$, $q_y = 0.5$), computes:

1. The best-fit line through the entire 800-point coupler curve.
2. The pointwise absolute residuals $|y_j - \hat{y}_j|$.
3. The cumulative arc length parameter $s_j = \sum_{k=1}^{j} \sqrt{(\Delta x_k)^2 + (\Delta y_k)^2}$.

The result is plotted as deviation vs. arc length, with a dashed horizontal line at the 1% tolerance level (0.01 in normalised units). The region of the curve where the deviation stays below this line is the usable operating range for a seismometer.

### Section 11 – Symbolic Specialisation and Verification

Defines the symbolic coupler trace equation for the Roberts linkage by:

1. Setting $Q = \frac{1}{2} + iq_y$, $Q^* = \frac{1}{2} - iq_y$ (Roberts symmetry condition $Q^* = 1 - Q$).
2. Setting $A = A^* = 0$, $B = B^* = 1$ (normalised ground pivots).
3. Setting $l_1 = l_3 = \ell$ (Roberts length condition).
4. Computing $g$ and $h$ from eqs. (22)–(23).
5. Building the $4 \times 4$ matrix $M$ and computing `Det[M]` symbolically.

The output confirms that the result is degree 6 in $X$ and degree 6 in $X^*$, consistent with the sextic theorem. The number of monomial terms is also printed, allowing comparison with the 16-term structure discussed in Section 3.4.

### 8.3 Running the Notebook

1. Open `RobertsLinkage.nb` in Mathematica (version 12 or later).
2. Evaluate all cells in Section 2 first (`Shift+Enter` on each, or `Kernel > Evaluate Notebook`).
3. The `Manipulate` blocks in Sections 4 and 6 will render interactive sliders immediately after evaluation.
4. Section 7 (the deviation grid) takes approximately 30 seconds to compute.
5. Section 8 (`NMinimize`) takes 1–3 minutes depending on hardware.
6. Section 11 (symbolic determinant) may take several minutes; it is the most computationally expensive cell.

> **Tip.** If Mathematica is slow on the `Manipulate` blocks, reduce the `npts` argument in `RobertsTrace` from 800 to 300–400. The interactive response will be faster with minimal loss of visual quality.

## Summary of Key Equations

| Equation | Expression | Significance |
|---|---|---|
| (10a) | $x^2 + y^2 - 2(ux + vy)\cos\theta + 2(vx - uy)\sin\theta + u^2 + v^2 - a_2^2 = 0$ | Crank constraint |
| (10b) | (quadratic in $x, y$, linear in $\cos\theta, \sin\theta$) | Follower constraint |
| (10c) | $(10a) - (10b)$: linear in $x, y$ | Reduced constraint |
| (11) | $T = \tan(\theta/2)$, $\cos\theta = (1 - T^2)/(1 + T^2)$, $\sin\theta = 2T/(1 + T^2)$ | Weierstrass sub. |
| (12a,b) | $A_k T^2 - 2B_k T + C_k = 0$, $k = 1, 2$ | Quadratic in $T$ |
| (13a,b) | $M\mathbf{x} = \mathbf{0}$, $\mathbf{x} = (T^3, T^2, T, 1)^T$ | Linear system |
| (14) | $F(x, y) = \det(M) = 0$ | Coupler curve (sextic) |
| (5) | $4 \times 4$ determinant in $X, X^*$ | Coupler trace (isotropic |
| (9) | $f(X_m, X_m^*; \boldsymbol{d}) = \lambda f(X, X^*; \boldsymbol{d})$ | Symmetry condition |
| (15) | $\min \sum_j \eta_j^2$ | Optimisation objective |